

Industrial Safety Wearable: System Design Document

System Overview:

This document outlines the design and feature set for an industrial safety wearable utilizing an ESP32 microcontroller and a comprehensive array of biometric, environmental, and motion sensors. Combining the processing power of the ESP32 with this specific sensor array allows for real-time monitoring of a worker's physical health, immediate environment, and movements.

Below is a breakdown of the on-device features, display structuring, and central dashboard capabilities.

1. Wearable Device Features (On-Worker)

- **Fall and Impact Detection:** The MPU6050 (Accelerometer/Gyroscope) can detect sudden changes in acceleration and orientation, triggering an automatic alert if a worker falls or is struck.
- **Environmental Hazard Detection:** The MQ-135/MQ-2 continuously monitors for dangerous gas leaks, smoke, or poor air quality.
- **Heat Stroke Prevention:** The DHT11/DHT22 tracks surrounding temperature and humidity, alerting the worker if the heat index reaches unsafe levels.
- **Vitals Monitoring:** The MAX30102 reads heart rate (BPM) and blood oxygen levels (SpO2), identifying if a worker is experiencing extreme physical stress or oxygen deprivation.
- **Acoustic Trauma Warning:** The KY-038/MAX4466 sound sensor monitors decibel levels, logging prolonged exposure to loud machinery or detecting sudden explosive noises.
- **Elevation and Fall Context:** The BMP180 pressure sensor tracks altitude. Sudden pressure changes can corroborate a fall, and altitude data helps identify if a worker is operating at dangerous heights.
- **Emergency SOS:** The TTP223 touch button acts as a silent or audible panic alarm that the worker can tap in an emergency.
- **Multi-Sensory Alerts:** In noisy or visually distracting environments, the device uses the Buzzer (audio), LED (visual), and Coin Vibration Motor (haptic) simultaneously to ensure the worker notices danger.
- **Worker Identification:** The RFID module allows the worker to "tap in" with an ID card, assigning the device's data stream to their specific profile for the shift.
- **Black Box Data Logging:** If Wi-Fi drops, the ESP32 can save all sensor telemetry to the SD Card Module, syncing it to the server once the connection is restored.
- **Power Management:** The 18650 battery, charged via the TP4056 and regulated by the MT3608 boost converter, ensures a steady power supply for a full shift.

2. OLED Display Features (0.96-inch)

Because a 0.96-inch display is extremely small, clutter must be avoided. The ESP32 is programmed to rotate through different "pages" every few seconds, or uses the TTP223 touch button to cycle through the following screens:

- **Main Status Screen:** Shows current Time, Wi-Fi signal strength, Battery percentage, and the assigned Worker ID (read from the RFID).
- **Vitals Screen:** Displays a heart icon with the current BPM and an oxygen icon with the SpO2 percentage.
- **Environment Screen:** Shows current Temperature (°C/°F), Humidity (%), and an "Air Quality: OK/BAD" status.
- **Emergency Override Screen:** If a threshold is breached (e.g., gas detected, fall detected), the display immediately switches to a flashing, high-contrast text alert (e.g., "EVACUATE" or "GAS ALERT") to accompany the buzzer and vibration.

3. Online Dashboard Features (Manager/Admin View)

The ESP32 transmits data over Wi-Fi to a central server. This web dashboard acts as a central command center for site safety managers.

- **Live Operations Map:** A grid or list showing all active workers, their current connection status, and their last known zone (inferred by which Wi-Fi router they are connected to or RFID checkpoints).
- **Real-Time Telemetry Feed:** A constantly updating table showing live BPM, SpO2, surrounding temperature, and air quality for every active wearable.
- **Critical Alert Center:** A priority notification panel that flashes red and sounds an alarm on the manager's computer if a Fall, SOS button press, or Gas Leak is detected on any device.
- **Historical Exposure Logs:** Downloadable reports showing a worker's daily exposure to heat, noise, and poor air quality, useful for OSHA compliance and health audits.
- **Device Management:** A page tracking the battery levels of all wearables on the floor, alerting the manager if a device goes offline or drops below 10% battery.
- **User Profiles:** A database linking RFID tag numbers to specific worker names, emergency contact information, and medical notes.